LESSON 8.5

Understand Inequalities and Their Solutions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can graph the solutions of an inequality on a number line.

WHY THIS MATTERS Detective Okafor is building a timeline. The suspect was seen at the scene no earlier than 2:00 PM, meaning the arrival time t satisfies $t \geq 2$. She needs to plot this on a timeline to share with her team. How should she mark the number line?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Number line *Spanish: Recta numérica*	A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right.	A horizontal line with marks at 0, 1, 2, 3, ... showing where values fall
Open circle *Spanish: Círculo abierto*	An empty circle showing a number is NOT included.	$x > 5$: open circle at 5 means 5 itself is NOT a solution, but 5.1, 6, 7, ... are
Closed circle *Spanish: Círculo cerrado*	A filled-in circle showing a number IS included.	used for $\leq$ and $\geq$
Solution set *Spanish: Conjunto solución*	All the numbers that make the inequality true.	every number 8 or above
Inequality *Spanish: Desigualdad*	A math sentence that compares two sides with <, >, $\leq$, or $\geq$.	$x \geq 3$ is an inequality; it is graphed as a closed circle on 3 with an arrow to the right.
Boundary point *Spanish: Punto límite*	The number on the number line where the solution set starts.	the 8 in $x \geq 8$

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Graph $x < 6$ on a number line.

1. The boundary number is 6.
2. < does not include 6, so draw an open dot at 6.
3. Less than 6 means the arrow points left. Test: $5 < 6$ is true.

Answer: open dot at 6, arrow pointing left

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Graph x < 5. What number is the boundary?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- My inequality is _____ because the phrase says _____ .
- I drew a _____ dot at _____ because _____ .
- My arrow points _____ because the solutions are _____ .
- The value _____ sits under my arrow, so it _____ .

Word bank inequality | graph | number line | boundary point | closed circle | open circle | arrow | shade | solution set | at least | at most | include

WATCH OUT!

A common mistake

A common mistake in Graph Inequalities is treating ≥ and ≤ the same as > and < and drawing the wrong circle. For example, students often graph x ≥ 6 with an OPEN circle (as if it were x > 6), which wrongly excludes 6 even though 6 itself makes x ≥ 6 true. A second common mistake is choosing the right circle but shading the wrong direction

LESSON 8.5

Understand Inequalities and Their Solutions

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x < 5$?

(A) Open circle at 5, shaded left
(B) Closed circle at 5, shaded left
(C) Open circle at 5, shaded right
(D) Closed circle at 5, shaded right

STEPS TO TRY

1. Every inequality graph asks two questions: is 5 itself a solution (circle type), and which side holds the solutions (shading)?
2. Strict symbols like < exclude the boundary, so check whether 5 makes $x < 5$ true. Then shade toward test values that work.
3. First step: $5 < 5$ is false, so 5 is NOT included — that fixes the circle type. Now test $x = 4$ to find the shading direction.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x \geq 3$?

(A) Closed circle at 3, shaded right

(B) Open circle at 3, shaded right

(C) Closed circle at 3, shaded left

(D) Open circle at 3, shaded left

STEPS TO TRY

1. Check the symbol: ≥ has the 'or equal' bar. What does that bar mean for the circle at 3?
2. The bar means 3 IS a solution, so the circle gets filled in. Then a quick test value tells you the shading side.
3. First step: $3 \geq 3$ is true, so 3 is included. Now test $x = 10$ — is $10 \geq 3$ true? Shade toward that side.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x > 8$?

(A) Open circle at 8, shaded right

(B) Closed circle at 8, shaded right

(C) Open circle at 8, shaded left

(D) Closed circle at 8, shaded left

STEPS TO TRY

1. Two decisions: does $x = 8$ make $x > 8$ true, and are the solutions bigger or smaller than 8?
2. Symbols without the 'or equal' bar exclude the boundary. Then shade only the side where test numbers make the statement true.
3. First step: $8 > 8$ is false, so 8 is NOT included — that fixes your circle. Now test $x = 9$ to pick the shading side.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x \leq 10$?

(A) Closed circle at 10, shaded left

(B) Open circle at 10, shaded left

(C) Closed circle at 10, shaded right

(D) Open circle at 10, shaded right

STEPS TO TRY

1. Ask two things: does 10 itself satisfy $x \leq 10$, and do solutions live above or below 10?
2. The 'or equal' bar means the boundary is a solution — that decides the circle. A test value decides the shading.
3. First step: $10 \leq 10$ is true, so 10 is included. Now test $x = 3$ — is $3 \leq 10$ true? Shade toward the side that works.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide between an open circle and a closed circle when graphing?

SENTENCE STARTER I use a ___ circle when the symbol is ___ because the boundary value ___ part of the solution.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help